# Solubility and Colligative Properties

Answer key with worked solutions

## A. Understanding

**1.** Definitions.

> (a) A property that depends only on the **number** of dissolved particles, not their identity. (b) Moles of solute per **kilogram of solvent** (mol/kg). (c) The number of particles each formula unit gives in solution. (d) Dissolved particles lower the **vapour pressure** of the solvent (fewer solvent molecules escape from the surface), so a higher temperature is needed for the vapour pressure to reach atmospheric pressure.

**2.** Solubility.

> (a) Sugar has many –OH groups and forms **hydrogen bonds** with water; hexane is non-polar and cannot attract it (like dissolves like). (b) **Gases are less soluble** in warm water, so there is less dissolved oxygen. (c) The drink is bottled under high $CO_2$ pressure; opening it **lowers the pressure**, so by Henry's law less $CO_2$ can stay dissolved and it forms bubbles.

## B. Freezing and boiling points

**3.** Glucose.

> (a) $n = \dfrac{18.0\ \text{g}}{180.16\ \text{g/mol}} = 0.09991\ \text{mol}$; $m = \dfrac{0.09991\ \text{mol}}{0.100\ \text{kg}} = \mathbf{0.999\ mol/kg}$
>
> (b) $\Delta T_f = 1 \times 1.86\ \frac{^\circ\text{C kg}}{\text{mol}} \times 0.999\ \frac{\text{mol}}{\text{kg}} = 1.86\ ^\circ\text{C}$, so it freezes at **−1.86 °C**
>
> (c) $\Delta T_b = 1 \times 0.512\ \frac{^\circ\text{C kg}}{\text{mol}} \times 0.999\ \frac{\text{mol}}{\text{kg}} = 0.512\ ^\circ\text{C}$, so it boils at **100.512 °C**

**4.** Order of freezing points.

> Highest first: **glucose** (i = 1, 0.10 mol/kg of particles), **NaCl** (i = 2, 0.20 mol/kg), $CaCl_2$ (i = 3, 0.30 mol/kg). More particles give a larger depression, so a lower freezing point.

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**5.** Calcium chloride.

> $n = \dfrac{11.1\ \text{g}}{110.98\ \text{g/mol}} = 0.1000\ \text{mol}$; $m = \dfrac{0.1000\ \text{mol}}{0.250\ \text{kg}} = 0.4001\ \text{mol/kg}$
>
> $\Delta T_f = 3 \times 1.86\ \frac{^\circ\text{C kg}}{\text{mol}} \times 0.4001\ \frac{\text{mol}}{\text{kg}} = 2.23\ ^\circ\text{C}$, so it freezes at **−2.23 °C**

**6.** Molar mass from a boiling point.

> $m = \dfrac{\Delta T_b}{iK_b} = \dfrac{0.256\ ^\circ\text{C}}{1 \times 0.512\ ^\circ\text{C kg/mol}} = 0.500\ \text{mol/kg}$
>
> $n = 0.500\ \text{mol/kg} \times 0.0500\ \text{kg} = 0.0250\ \text{mol}$; $M = \dfrac{3.00\ \text{g}}{0.0250\ \text{mol}} = \mathbf{120.\ g/mol}$

## C. Osmotic pressure

**7.** Protein.

> $\Pi = \dfrac{3.80\ \text{torr}}{760\ \text{torr/atm}} = 5.00 \times 10^{-3}\ \text{atm}$; $T = 298.15\ \text{K}$
>
> $M_{\text{conc}} = \dfrac{\Pi}{RT} = \dfrac{5.00 \times 10^{-3}\ \text{atm}}{(0.08206\ \text{L atm mol}^{-1}\,\text{K}^{-1})(298.15\ \text{K})} = 2.044 \times 10^{-4}\ \text{mol/L}$
>
> $n = 2.044 \times 10^{-4}\ \text{mol/L} \times 0.1000\ \text{L} = 2.044 \times 10^{-5}\ \text{mol}$; molar mass $= \dfrac{1.00\ \text{g}}{2.044 \times 10^{-5}\ \text{mol}} =$ **$4.89 \times 10^4$ g/mol**

**8.** Isotonic saline.

> $T = 37.0 + 273.15 = 310.15\ \text{K}$
>
> $M = \dfrac{\Pi}{iRT} = \dfrac{7.70\ \text{atm}}{2 \times (0.08206\ \text{L atm mol}^{-1}\,\text{K}^{-1})(310.15\ \text{K})} = \mathbf{0.151\ mol/L}$

## D. Challenge

**9.** Road salt.

> $m = \dfrac{\Delta T_f}{iK_f} = \dfrac{10.0\ ^\circ\text{C}}{2 \times 1.86\ ^\circ\text{C kg/mol}} = 2.688\ \text{mol/kg}$, so 2.688 mol in 1.00 kg of water
>
> mass $= 2.688\ \text{mol} \times 58.44\ \text{g/mol} =$ **157 g** of NaCl
>
> $CaCl_2$ gives **three** ions per formula unit (i = 3) instead of two, so it lowers the freezing point more per mole, and it can lower the freezing point much further (it is very soluble).

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